

API

(Application Programming Interface)

Mobile App Development

What is an API?

- API are tool that allows different software or applications to communicate with each other.
- Enable them to share data and functionality.
- Flutter App → API → Server/Database

Why are APIs Important?

- Fetch data
- Send data
- Use third-party services
- APIs make development faster by connecting services.

Types of APIs (Access Based)

- Open API
- Private API
- Partner API
- Composite API

Open API

- Publicly available
- Used for learning & public data
- **Example:** Weather, News APIs

Private API

- Used within organizations
- Secure internal data
- **Example:** University or Banking systems

Partner API

- Shared with selected partners
- Used for business collaboration
- **Example:** Payment gateways

Composite API

- Combines multiple APIs
- Improves performance
- Used in complex apps

Types of APIs (Architecture)

- REST API
- SOAP API
- GraphQL API
- WebSocket API

REST API

(Representational State Transfer)

- Most common web architecture based API
- Uses HTTP methods

When to use it?

- When you need a simple stateless API
- Best for CRUD operations used in Flutter apps

SOAP API

(Simple Object Access Protocol)

- A protocol for exchanging structured information
- Uses XML
- Follow the Strict rules

When to use it?

- Used it when high security are required
- Used in enterprise application. (financial services, payment gateways)

GraphQL API

- Single endpoint
- Client requests specific data
- Used in modern systems

When to use it?

- When you need more control over the data you fetch.
- Great for application when complex data relationships.

WebSocket API

- Enable real-time, two-way communication between client and server.

When to use it?

- When you need real-time communication, like chat applications or live updates.

HTTP Methods

- GET – Fetch data
- POST – Send data
- PUT – Update data
- DELETE – Remove data

How API Works in Flutter App

- App sends request
- API processes request
- Server returns JSON
- Data shown on UI